

Roll No.

TEC-201

B. Tech End Semester Examination 2025
2nd Semester
Basic Electronics Engineering

Time: Three Hours

MM: 100

Note:

- (i) All questions are compulsory.
(ii) Answer any two sub questions among a, b & c in each main question.
(iii) Total marks for each main question is twenty.
(iv) Each question carries 10 marks.

Q1	(20 marks)	CO1
(a)	Convert the following bases (i) $(4FB2)_{16} = (?)_2 = (?)_8 = (?)_{10}$ (ii) $(3479)_{10} = (?)_2 = (?)_8 = (?)_{16}$	
(b)	State and prove De Morgan theorem. Prove the following using Boolean Algebra: $(A+B)(A+B')(A'+C) = AC$	
(c)	Solve the following expression using K – Map and relase the output with NAND gate only. $Y(A, B, C, D) = \sum m(0, 2, 4, 6, 7, 8, 10)$	
Q2	(20 marks)	CO2
(a)	Based on energy band diagram explain conductor, semiconductor and insulator. Also, explain intrinsic and extrinsic semiconductor.	
(b)	(i) Draw the circuit diagram of diode in forward and reverse bias. Also draw the complete V-I characteristic of p-n junction diode. (ii) Determine the diode current at room temperature for Si diode with reverse saturation current 50 nA and applied forward voltage of 0.6 volt.	
(c)	Explain mass action law. Find the conductivity of intrinsic Si at 300 k. It is given that intrinsic concentration is $1.5 \times 10^{10} / \text{cm}^3$ and the mobility of electrons and holes in Si are $1300 \text{ cm}^2/\text{V-s}$ and $500 \text{ cm}^2/\text{V-s}$ respectively. If donor type impurity is added to the extent of 1 impurity atom in 10^8 Si atoms then find its conductivity.	
Q3	(20 marks)	CO2, CO1
(a)	Draw the circuit diagram of Bipolar Junction Transistor (BJT) in Common Emitter (CE) mode for n-p-n transistor. Explain its input/output characteristics by mentioning the saturation, active and cutoff mode.	
(b)	Define α and β in bipolar junction transistor? Determine the relationship between α and β . In a certain transistor collector current, I_C , is 0.98 mA and base current, I_B , is 20 μA . Determine the values of emitter current, α and β .	
(c)	Determine the base in following equations: (i) $\frac{14}{2} = 5$ (ii) $(211)_x = (152)_8$	

Q4	(20 marks)	CO3
(a)	Explain half wave rectifier with diagram and input/output waveforms. Determine the efficiency and ripple factor for the half wave rectifier.	
(b)	With proper diagram explain the working of Full Wave Center Tap and Bridge Rectifier. Also define the term Peak Inverse Voltage (PIV) and provide its value in case of Center Tap and Bridge Rectifier.	
(c)	A diode having internal resistance of R_f is 20Ω used for half wave rectification. If the applied voltage $v=50\sin\omega t$ and load resistance R_L is 800Ω . Determine the I_{max} , I_{dc} , I_{rms} , P_{ac} , P_{dc} .	
Q5	(20 marks)	CO4
(a)	Write down the characteristics for ideal op-amp and draw the schematic of inverting and non-inverting configuration.	
(b)	Design an Adder circuit using an op-amp to get the output expression as $V_{out} = -(V_1 + 10 V_2 + 100 V_3)$ where V_1 , V_2 and V_3 are the inputs. Given that $R_f=100k\Omega$	
(c)	Solve the following expression using K-Map and realize the output using NOR gate only: $F(A,B,C,D) = \sum m(1, 5, 6, 11, 12, 15) + \sum d(7, 13)$	